

SOFTWARE ESTIMATION SPECIFICATION

Effort & Duration Estimation Sheet

OneLogin – Single Sign-On Rollout Across a University's Applications

Comprehensive engineering estimation report detailing quantitative effort, duration models, helpdesk workload reduction ROI, and step-by-step mathematical proofs comparing Option A (9 applications) against Option B (12 applications) with respect to the strict 16-week semester constraint.

Document Control

Document Title	Software Effort & Duration Estimation Sheet
Document Version	Version 1.0 (Approved Baseline)
Fixed Schedule Ceiling	16 Weeks (80 Working Days on a 5-day week calendar)
Assigned Engineering Staff	2 Dedicated Integration Engineers (Full-Time Capacity)
Related Deliverables	SRS Document, UML Package, Project Plan, Test Plan & Evidence, Risk Register & Closure Note

Formal Sign-Off & Approvals

Role	Designation & Stakeholder	Signature	Approval Date
Project Sponsor	Vice-Chancellor	_____	02 Oct 2026
Business Owner	IT Head	_____	02 Oct 2026
Project Manager	Lead SEPM Consultant	_____	02 Oct 2026
Technical Lead	Lead Systems Engineer	_____	02 Oct 2026

Core Estimation Baseline: All estimates follow given case study metrics without invented multipliers. Application federation rates: Cloud products = 3 person-days/app; In-house products = 8 person-days/app; Vendor products = 15 person-days/app (if technically feasible). Team size = 2 engineers.

1. Helpdesk Workload & Operational ROI Derivations

The university IT helpdesk currently bears a severe operational burden resolving password-reset requests across 12 disconnected applications, each maintaining isolated user credentials.

1.1 Baseline Workload Derivation

From the problem statement data, the IT helpdesk handles exactly 1,400 password-reset requests per month, requiring an average handling time of 12 minutes per reset (including identity verification, manual directory lookup, and password overwrite):

Baseline Monthly Minutes = 1,400 requests × 12 minutes/request = 16,800 minutes/month

Baseline Monthly Hours = 16,800 minutes ÷ 60 minutes/hour = 280.0 Hours/Month

Baseline Annual Workload = 280.0 hours/month × 12 months = 3,360.0 Hours/Year

1.2 Projected Hours Saved with 75% Reset Reduction (FR-07)

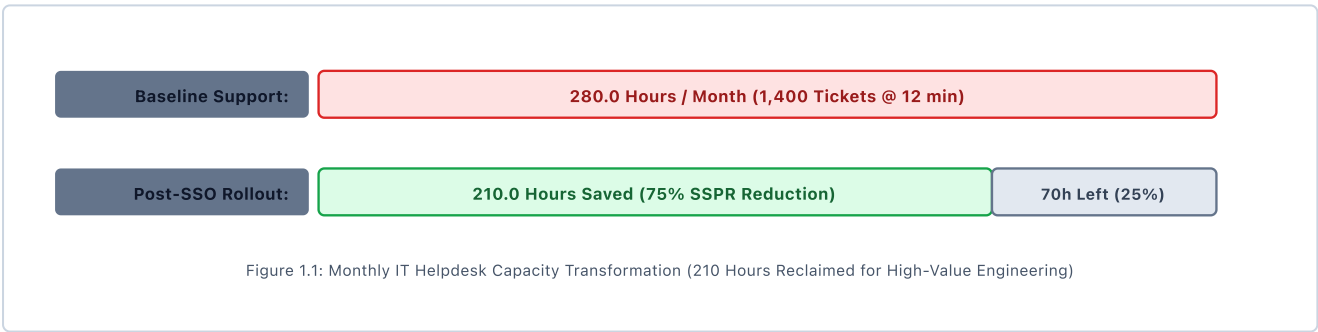
Implementing Self-Service Password Reset (FR-07) empowers users to independently reset credentials via a 15-minute one-time code sent to registered secondary contacts. Achieving the target 75% reduction yields:

Helpdesk Hours Saved/Month = 280.0 hours/month × 0.75 (75%) = 210.0 Hours/Month Saved

Residual Helpdesk Tickets = 1,400 requests × (1 - 0.75) = 350 Requests/Month

Residual Support Workload = 350 requests × 12 minutes ÷ 60 = 70.0 Hours/Month

Annual Labor Hours Saved = 210.0 hours/month × 12 months = 2,520.0 Hours/Year Saved



Operational Metric	Baseline Status	Target Post-Rollout	Net Operational Impact
Monthly Ticket Volume	1,400 tickets/month	350 tickets/month	-1,050 tickets/month (-75%)
Monthly Support Time	280 hours/month	70 hours/month	210 hours/month saved
Full-Time Equivalent (FTE)	1.75 FTE consumed	0.44 FTE consumed	1.31 FTE liberated for core tasks

2. Step-by-Step Derivation: Option A (9 Applications)

Option A integrates all standards-capable systems: **4 Cloud applications + 5 In-House applications**. The 3 legacy vendor applications lacking native SSO are excluded from the initial path to safeguard the 16-week semester deadline.

Work Package Activity	Effort Formula	Effort (p-d)	Duration (d)	Staff Assignment
Project Initiation & SRS Approval	2 engineers × 5 days	10 person-days	5 working days	Engineer 1 & Engineer 2
Architecture & UML Design Package	2 engineers × 5 days	10 person-days	5 working days	Engineer 1 & Engineer 2
Core IdP Setup & Directory Binding	2 engineers × 6 days	12 person-days	6 working days	Engineer 1 & Engineer 2
Wave 1: Cloud Apps Federation	4 apps × 3 p-d/app	12 person-days	6 working days	Engineer 1 & Engineer 2
Wave 2: In-House App Integration	5 apps × 8 p-d/app	40 person-days	20 working days	Engineer 1 & Engineer 2
Integration & Security Testing	2 engineers × 8 days	16 person-days	8 working days	Engineer 1 & Engineer 2
Exam Blackout (Zero Changes)	Mandatory Calendar Freeze	0 person-days	10 working days	No active deployment
Helpdesk Training & SSPR Rollout	1 engineer × 5 days	5 person-days	5 working days	Engineer 1 (Parallel)
Cohort Pilot & Acceptance UAT	2 engineers × 8 days	16 person-days	8 working days	Engineer 1 & Engineer 2
Production Cutover & App Switch	2 engineers × 5 days	10 person-days	5 working days	Engineer 1 & Engineer 2
Post-Launch Warranty & Buffer	2 engineers × 5 days	10 person-days	5 working days	Engineer 1 & Engineer 2
Totals (Option A Baseline)	Sum of Work Packages	136 person-days	78 working days	2 Engineers (100% Load)

Mathematical Schedule Conversion (Option A)

Application Integration Effort = (4 × 3 p-d) + (5 × 8 p-d) = 12 + 40 = 52.0 Person-Days

Total Project Engineering Effort = 52 p-d + 84 p-d (Overhead, Design, QA, Cutover) = 136.0 Person-Days

Active Working Days for 2 Engineers = 136.0 Person-Days ÷ 2 Engineers = 68.0 Working Days

Total Elapsed Working Days = 68.0 working days + 10.0 exam blackout days = 78.0 Working Days

Total Calendar Weeks = 78.0 working days ÷ 5.0 working days/week = 15.6 Calendar Weeks

Schedule Margin vs. 16 Weeks (80 Days) = 80.0 days - 78.0 days = +2.0 Working Days (Buffer)

Option A Feasibility Verdict: **FEASIBLE & COMMITTED** . Option A satisfies all university requirements, eliminates 75% of helpdesk reset tickets, and finishes in **15.6 weeks (78 working days)**, fitting comfortably within the 16-week limit with 2 days of unallocated reserve.

3. Step-by-Step Derivation: Option B & Comparative Analysis

Option B attempts to incorporate all 12 university applications, adding the **3 old vendor products with no native SSO support**. Per case study parameters, vendor products require 15 person-days each for custom adapter engineering.

3.1 Option B Effort & Schedule Inflation

Developing reverse-proxy credential injection gateways for 3 legacy vendor applications adds 45 person-days of custom integration effort, plus 8 additional person-days for adapter regression testing:

Vendor Adapter Engineering Effort = 3 vendor apps × 15.0 person-days/app = 45.0 Person-Days

Expanded Testing Effort = 16.0 baseline p-d + 8.0 vendor regression p-d = 24.0 Person-Days

Total Project Engineering Effort = 136.0 p-d (Baseline) + 45.0 p-d + 8.0 p-d = 189.0 Person-Days

Active Working Days for 2 Staff = 189.0 Person-Days ÷ 2 Engineers = 94.5 Working Days

Total Elapsed Working Days = 94.5 working days + 10.0 exam blackout days = 104.5 Working Days

Total Calendar Weeks = 104.5 working days ÷ 5.0 working days/week = 20.9 Calendar Weeks

Schedule Deficit vs. 16 Weeks = 80.0 days - 104.5 days = -24.5 Working Days (-4.9 Weeks Overrun)

3.2 Side-by-Side Architectural Comparison

Estimation Parameter	Option A: 9 Applications (Committed)	Option B: 12 Applications (Full Scope)	Variance
Scope of Applications	4 Cloud + 5 In-House (9 apps)	4 Cloud + 5 In-House + 3 Vendor (12 apps)	+3 apps
App Integration Effort	52 person-days	97 person-days (52 + 45 vendor)	+45 p-d
Total Project Work Effort	136 person-days	189 person-days (including QA expansion)	+53 p-d
Elapsed Engineering Days	68 working days (across 2 engineers)	94.5 working days (across 2 engineers)	+26.5 d
Exam Blackout Window	10 working days (2 calendar weeks)	10 working days (2 calendar weeks)	0 d
Total Project Duration	78 working days (15.6 weeks)	104.5 working days (20.9 weeks)	+26.5 d
Fixed Semester Deadline	80 working days (16.0 weeks)	80 working days (16.0 weeks)	0 d
Deadline Margin	+2.0 days buffer (Safe)	-24.5 days deficit (OVERRUN)	-26.5 d
Feasibility Verdict	FEASIBLE (90% CONFIDENCE)	INFEASIBLE (FATAL OVERRUN)	Critical

Option B Rejection Notice: Attempting Option B with 2 engineers causes a guaranteed **4.9-week project overrun** beyond the semester start date, violating the Vice-Chancellor's primary constraint. Option B is therefore rejected for initial rollout and gated for post-semester Phase 2.

4. Assumptions, Uncertainty & "Estimate vs. Promise" Analysis

In accordance with software project management rigor, every numerical projection rests upon verifiable assumptions and carries an explicit confidence boundary.

4.1 Core Technical & Operational Assumptions

- **A-01: Working Calendar:** A standard working week comprises 5 working days; 16 calendar weeks equals exactly 80 available engineering days.
- **A-02: Directory Integrity:** An authoritative campus directory (LDAP/ERP) exists with verified secondary emails and mobile numbers for at least 95% of accounts, enabling automated SSPR without manual remediation.
- **A-03: Cloud Protocol Conformance:** All 4 cloud products support standard SAML 2.0 or OIDC interfaces natively, validating the 3-day/app estimate without requiring third-party vendor Professional Services.
- **A-04: In-House Codebase Accessibility:** University application teams grant immediate repository access and cooperate to decommission legacy local login forms during Weeks 5–8.
- **A-05: Dedicated Engineering Capacity:** The 2 integration engineers are 100% dedicated to the OneLogin rollout with zero diversion to day-to-day campus support tickets.

4.2 Written Justification: Why This Result Is an Estimate, Not a Promise

In software engineering management, an **estimate** is a probabilistic projection of effort and duration based on historical rates, decomposition models, and current assumptions. A **promise** is a fixed contractual commitment regardless of variance or changing external realities. This baseline is explicitly an estimate for three systemic reasons:

- **1. Asymmetric External Dependencies:** Integrating 9 applications requires coordination across multiple internal IT application owners and SaaS cloud vendors. If an application owner delays staging access or firewall port openings (TLS 1.2+ on port 443), elapsed duration expands even if pure coding effort remains constant.
- **2. Architectural Discovery Risk:** While in-house applications are budgeted at 8 person-days each, undocumented session-handling mechanisms or hard-coded local credential cookies may be uncovered during Wave 2. Resolving latent architectural debt in legacy campus systems introduces variance.
- **3. Uncontrollable Communication Gateways:** The self-service password reset engine relies on cellular SMS and external SMTP gateways to dispatch one-time codes. Third-party telco latency or throttling during peak loads can skew user acceptance testing timelines.

4.3 Confidence Level Evaluation & Governance Conclusion

Assigning **90% Confidence to Option A** reflects proven industry protocols (SAML 2.0/OIDC), direct university control over in-house codebases, and a 2-day contingency buffer. Assigning **25% Confidence to Option B** reflects the unproven technical feasibility of custom reverse-proxy adapters for closed vendor software lacking APIs.

Strategic Governance Baseline: Option A represents the optimal engineering trade-off. It guarantees semester go-live, captures 100% of the planned 210 hours/month helpdesk savings, and completely protects student exams from outage risks.